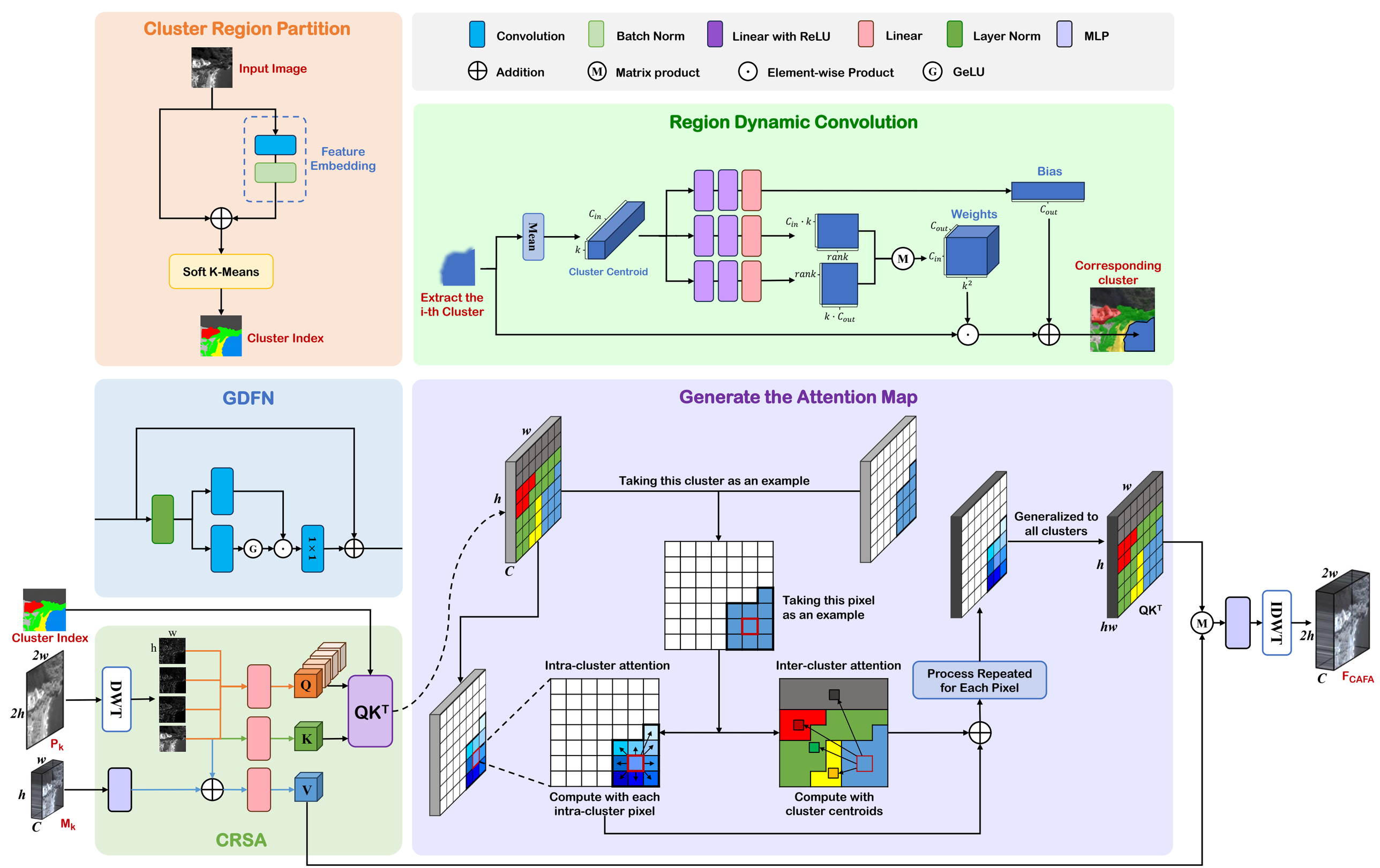
Cluster Region Partition
Input Image
Feature Embedding
Soft K-Means
Cluster Index
Convolution
Batch Norm
Linear with ReLU
Linear
Layer Norm
MLP
Addition
Matrix product
Element-wise Product
GeLU
Region Dynamic Convolution
Extract the i-th Cluster
Mean
Cluster Centroid
Bias
Weights
Corresponding cluster
GDFN
Generate the Attention Map
Taking this cluster as an example
Taking this pixel as an example
Generalized to all clusters
Intra-cluster attention
Inter-cluster attention
Process Repeated for Each Pixel
Compute with each intra-cluster pixel
Compute with cluster centroids
Cluster Index
DWT
IDWT
CRSA
QKT
Q
K
V
Pk
Mk
FCAFA